studocu

Health CARE Statistics Notes

health records and information technology (Kenya Medical Training College)

Scan to open on Studocu

Studocu is not sponsored or endorsed by any college or university
Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

HEALTH ADMINISTRATIVE STATISTICS

1. Explain health administrative statistics,importance and uses.
2. Describe procedure for collecting , compiling , computing and analyzing health administrative statistics
3. Interpret and present health administrative statistics

Assignment

1. In2001, ahospital unit recorded 11676 in patient days; 130 discharges and deaths and had 30 allocated beds. Calculate to 1 d.p

a) Available bed days
b) Excess / vacant bed days
c) Percentage occupancy
d) Average length of stay
e) Supposing this unit represents 25% of the entire hospital's beds patient and their days;
   i. Calculate the entire hospital statistics.State;
      -ABD
      -EBD/VBD
      -ALOS
      -% Occupancy

The meaning of (c)

2. I n 2001 Soliki hospital had allocated beds and cots. During the same year the hospital admitted 3144 patients with 3242 discharges and deaths. The totals in patient days for the year were 28600. Calculate the following values for use at Soliki hospital administration;

a) Available bed days
b) Excess / vacant bed days
c) Daily occupancy
d) Turn over interval
e) Turn over per bed
f) Length of stay

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

g) Explain your answer b, c, and e in relation to the use of available beds

3. In 2002, Makuenihospital had a bed complement of 120 and an average length of stay of 6 days. It also had a daily occupancy of 105 with a bed turnover of 54 patients. This turnover is expected to increase by 12% this year (2003). Using this data, calculate the following information for Makueni;

a) Available bed days
b) Occupied bed days
c) % occupancy
d) Discharges and deaths
e) Excess / vacant bed days
f) Turnover interval

4. In 1979, psychiatric unit recorded the following data. In – patients days 10766, number of allocated 25 through the year, discharges and deaths 124. Calculate
   ii. Excess in – patient days
   iii. The extra number of beds required to cater for excess patients days through the year
   iv. % occupancy
   v. Average length of stay
   vi. Average daily population

5. In 1990, a surgical department recorded thefollowing data . Allocated beds 56 , total occupied bed days 17062. 20 beds were closed from 1st June to 13th August inclusive .Total discharge 1921. Calculate
   a) Average occupied beds
   b) Average available
   c) % occupancy
   d) Average duration of stay
   e) Turnover interval
   f) Turnover per bed

Uses of health statistics

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Management of health services
- Health information aids planners in their day to day play
- Health activity analysis e.g. finance, how beds are used, staff and many other physical facilities.

Patients / clients

Patients / clients attendance make use of facilities in patient or outpatient

In patient statistics

Indicates the facilities of in-patient services in a health institution

The data iscollected through the form common known as daily bed return / bed state

Daily summary form

Summarizes all the activities

FORMULAS FOR COMPUTING HOSPITAL ADMINISTRATIVE STATISTICS

1. Available bed days =period(days) Allocated beds

   ABD = PAB

2. Occupied bed days / in patient days = Direct summary of DBR state

   OBD= Period daily IP days / daily occupancy

   OBD= P DIPD /DO OR

   OBD = ABD OR

   OBD =LOS D+D

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

3. %occupancy = occupied bed days / available bed days 100

   %OCC=100

4. Vacant bed days= available bed days –occupied bed days

   VBD=ABD-OBD (daily bed vacancy period)

5. Excess bed days =OBD-ABD (Daily excess bed state period)

   EBD=OBD-ABD

6. Length of stay = occupied bed days /discharges and death
   Duration of stay
   Average length of stay
   LOS=OBD/D+D
7. VBD + OBD = ABD
8. OBD = ABD- VBD
9. Average number of occupied beds / average number of beds
   OB =
10. Deaths and discharges
    D+D =
11. Available beds / staffed beds / average available beds
    AB =
12. Turnover per bed =discharges & deaths / number of available beds
    TOB =
13. Discharge and deaths = turnover per bed x available beds
    D +D = TOBXAB
14. Turnover interval = vacant bed days / number of deaths & discharges
    TOI =
15. VBD = TOI X D+D
16. D+D =

TERMINOLOGY (TERMS)

I. Allocated beds / Available beds / Staffed beds / bed compliment

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

-This is the number of beds constantly (permanently) allocated to a ward or hospital for treatment of patients.

II. In - patient

-Person who has gone all through admission procedures and is expected to occupy a bed in hospital ward at the in – patient departments at a specified occurrence or session.

III. Period – This is usually the available time in which a patient can be treated while admitted and is usually counted in days. The common period of reference is usually 365/366 which is one year. However, months or part of the year can be computed as half ayear or quarter year.

IV. Bed state – Refers to the number of patients occupying hospitalbeds during a given period of time.

V. In – patient days (IPD) – It refers to occupied bed days (OBD)
This is the total number of patients remaining or occupying a bed in the ward or hospital at each day.

VI. Percentage occupancy –This is the percentage of the actual in – patient days in comparison to the maximum available patient days as determined by the allocated beds capacity during a given period.

VII. Available bed days – Is the total number of bed days available of which the patient can occupy. Those particular beds in ward or hospital. This usually gives the available bed space for patient to use.

VIII. Turnover per bed / through put for bed / bed turnover - refers to the number of patients expected to have used each bed available in a particular ward or hospital during the specified period of time.

IX. Turnover interval – Is the average period of time in which beds stay vacant between twosuccessive patients' use of the same bed. This may also be said to be the period between a discharge and admission on the same bed in ward or hospital. It ii also the number of days beds lie vacant between two successive admissions

X. Length of stay – number a patient occupies a bed thus from admission time of a particular patient to the discharge time of the same patient.

XI. Occupied bed days – is the total number of patients remaining in the hospital / ward each day added together over the reference period

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

XII. Average occupancy/ average number of patients – This is the average number of patients in a hospital during a specified period

XIII. Amenity beds – is a bed provided in a single room or small ward for the time being required by any patient on medical grounds for which a patient is charged part of the cost.

XIV. Day case - are patients attending as a non – residence patient for investigations , therapeutic test, operative procedures or other treatment and who require some form of preparation period of recovery or both involving provision of accommodation and services

HEALTH ADMINISTRATIVE STATISTICS

Daily bed return

It's a tool of collection of data in a health facility offering admissions.

Information on DBR – admission, deaths, discharges, IP number , number of patients, transfers, occupation, well people , paroles , abscondees , name of patients , ward number , name of institution, compilation officer, balancing officer , previous state , date of completion.

Data source department – Availability – responsibility of HR&IT Department.

- ✓ Issued to the compiling department / section
- ✓ Balancing in the ward. The balancing nursing office or m.o.d in the ward
- ✓ Collection / assembling - sometimes collected by HR &I officer , sometimes delivered by the nursing department
- ✓ Balancing in the HR&I office

  Todays' daily bed state is equal to the previous daily state plus all admissions minus all discharges.

  *prepare an entry sheet.

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Statistics

1) Define the term statistics (4 mks)

2) Why should health records and information students learn statistics (8 mks)

3) State 8 uses of statistics (16 mks)

4) State the four types of statistics (8 mks)

MEDIAN

Denoting or relating to a value or quantity lying at the midpoint of a frequency distribution of observed values or quantities, such that there is an equal probability of falling above or below it

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

To find the **Median**, place the numbers you are given in value order and find the middle number. Example: find the **Median** of {13, 23, 11, 16, 15, 10, 26}. Put them in order: {10, 11, 13, 15, 16, 23, 26} The middle number is 15, so the **median** is 15. (If there are two middle numbers, you average them.)

ODD DISTRIBUTIONS

The following data was collected on age of patients attending a paediatric clinic

Step 1: 55,65,49,77,80,90,55,66,65,70,69

Step 2: 49,55,55,65,65,66,69,70,77,80,90

Median position = N+I/2 =11+1/2 =12/2 =6TH

Median position value is 66

Example 2

The following refers to the weights of the HR&I 2002 diploma class students born in the month of the first 3rd of a year (Jan ,Feb, Mar &Apr)

92,71, 60, 70, 75, 71, 50, 74 ,60 ,74 ,69 ,63, 62, 50, 71, 80 ,80

50 50 60 60 62 63 69 70 71 71 71 74 74 75 80 80 90

Median position = N+1/2 =17+1/2 =18/2 =9th position

Median position value is 71kg

INTERPRETATION OF MEDIAN

- ✓ Half of the diploma 2002 class students born in 1st 3rd year weighed 71kgs and above or
- ✓ Among the students weighed in the diploma HR&I 2002 half OR 50% weighed below 71kgs.

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

MEDIAN IN GROUPED DATA

To obtain median in grouped frequency distribution some of principles identified in the ungrouped data computations are observed ie arranging of the data in order of magnitude from the smallest to the highest and identifying the median position

However , there are two common methods followed in determining median among grouped frequency distribution , and this includes;

a) Calculation method
b) Diagrammatic method

CALCULATION METHOD

It is necessary to remember common characteristics in grouped frequency distribution to facilitate computation. This include

- Classes
- Frequency
- Class interval
- Class limit

Procedures

a. Obtain data and ensure that its grouped and in order of classes and with their sizes.
b. Workout the cumulative frequencies in a separate column usually abbreviated as CF (cf). Once you have worked out CT
c. Identify the median position by dividing the total frequency by two.Usually abb. As N/2(some abb it as ∑f/2)
d. Identify the location of median position in the distribution using the cumulative totals .

   The class in which the median position falls is referred to as the median class
e. Take note of the following identified median class

   i. The lower class limit abbreviated as L_1
   ii. Cumulative total level of the class before the median class abbreviated as CT2

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

iii. The frequency of the median abbreviated as fx

iv. Class interval especially as in the median class abbreviated as (i)

Obtain median by substitution of the values in number 5 in the following formula

Median = lower class limit + median position - cumulative total before median class x class interval /frequency of the median class

Median = L_1

Example 1

This sample data was collected from Namilama division during a research process on the role of the community on HIV/ AIDS awareness.

AGE	TOTALS(F)	Cf
15-19	40	40
20-24	130	170
25-29	89	259
30-34	64	323
35-39	30	353
40-44	22	375
45-49	14	389
50 yrs and above	16	405
Totals	405	

i. **Fin**d the median position

ii. Calculate the median

Interpretation

The total number of the half of the male and female between age of 15 and 50 years and above inNamilama division that wasinterviewed duringa research process on the role of the community on HIV/AIDS awareness was 26 years.

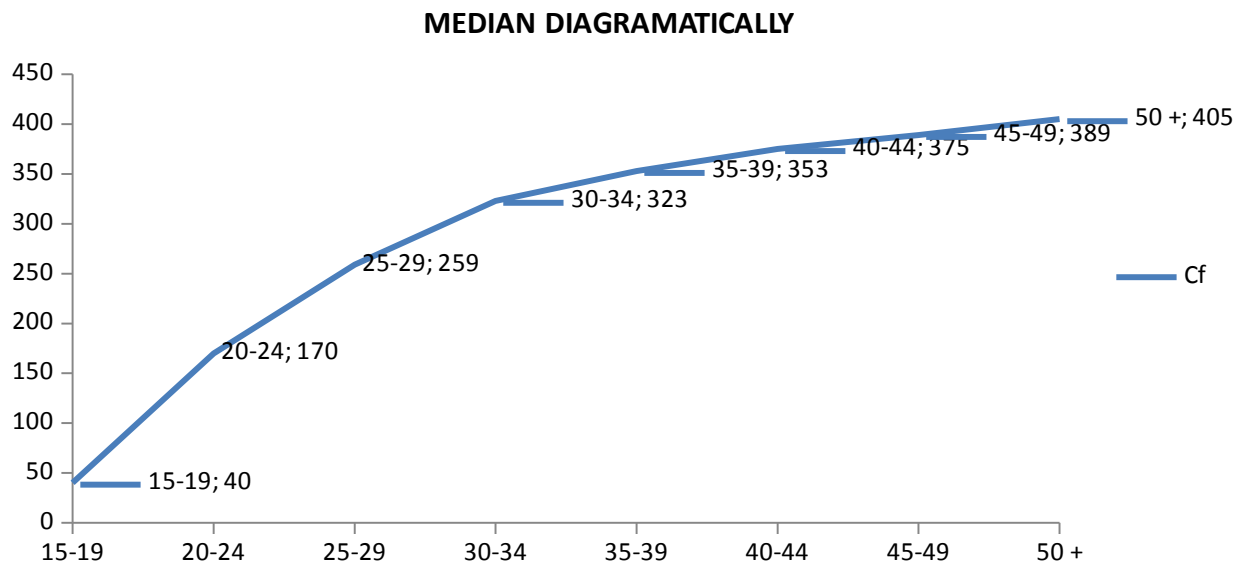
.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

MEDIAN DIAGRAMATICALLY

The principles acquired in cumulative frequency curve have to be ;

1) Draw / prepare a cumulative frequency curve.
2) Obtain the median position by dividing the total number of items (sum of frequency)by two abbreviated as N/2
3) Determine the median position value on the cumulative frequency curve
4) Trace the median value on the curve x-axis by drawing a perpendicular line on the cumulative frequency
5) The point of intersection of the perpendicular line with the x-axis is the median of the distribution . so determine its value



The median position is 26 years

Assignment

The following were on admission atolkalau nyayo ward last year

32,63,33,57,37,54,38,53,42,51,48,28,28,28,26,13,42,34,55,61,23,22,24,21,27,17,25,23,23,22,21,17,13,30,14,29,16,42,51,42,53,38,54,35,57,33,63,32,43,21,24,22,23,61,55,34,42,13,28,28,48,28,16,29,14,30,13,17,21,22,23,23,25,21,43,46,61,53,25,25,22,22,33,57,25,23,35,54,38,28,48,42,16,26,28,14,30

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

a) Using a class interval of 10 prepare a grouped frequency distribution table for the ages of the patient admitted in the ward (13marks).
b) Calculate the arithmetic mean from the table (5mks)
c) Find the median by calculation and diagrammatically from the table (10mks)
d) Give two portions of information arising from each of your answers in a-c above.

MODE

Is the measure of central tendency that represents a value or item ora observation that occurs most frequently . It is also a representative in a distribution or a series of figures which has the highest frequency .Mode is usually abbreviated as (z)

To identify mode in your distribution it is required that one identifies the particular type of data that is being handled ie ungrouped or grouped frequency distribution

MODE IN UNGROUPED DATA

Mode us obtained by inspection of the distribution to determine the item or observation that has occurred more than the others with the highest number of occurrence

Procedure

i. Must obtain data
ii. Inspect the data by looking at each item and determining numbers of times it occurs
iii. Select the item that has occurred most and that would be the mode for that particular distribution

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Example 1

Using data on diploma 2002 students weights on those born in the month of 1st quarter of the year were
50,92,71,60,70,75,71,74,60,74,69,63,62,50,71,80,80……

.STEPS

Arrange the values from the smallest to the highest ie 50,51,52……
92.Then you tally the values

Mode is 71kgs for the above data

Interpretation

Most of the students in the diploma class born in the month of the 1stquarter of the year were weighing 71kgs

MODE IN GROUPED DATA

Determination of mode in grouped data requires much more attention and it may follow two possible ways namely;

- Computation
- Diagrammatic presentation

However, both of them are worked from frequency distribution

COMPUTATION OF MODE

i. You obtain the frequency distribution table
ii. Inspect the table on the frequency column to determine the class with highest frequency
iii. The class with the highest frequency is usually referred to as modal class in a distribution
iv. The modal class is said to be carrying the modal value for the distribution
v. On the modal class , take note of the following
    - The lower class limit abbreviated as L_1
    - The frequency of the modal class abbreviated as f_1

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- The frequency of the item below or before the modal class abbreviated as f_o
- Frequency of the item above / after the modal class abbreviated as f_2
- Class interval of the distribution abbreviated as i

vi. Substitute the identified values of the modal to obtain mode. By using the following formula

Mode z =

Example 1

Using data from Namilama on the role of community on HIV / AIDS

Age	No of respondents
15-19	40
20-24	130
25-29	89
30-34	64
35-39	30
40-44	22
45-40	14
50+	16
Totals	405

Modal class =20-24

L_1 = 20

F_1 = 130

F_0 = 40

F_2 = 89

I = 5

Mode z =

=

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

=

=20+0.6875

=20+3.435

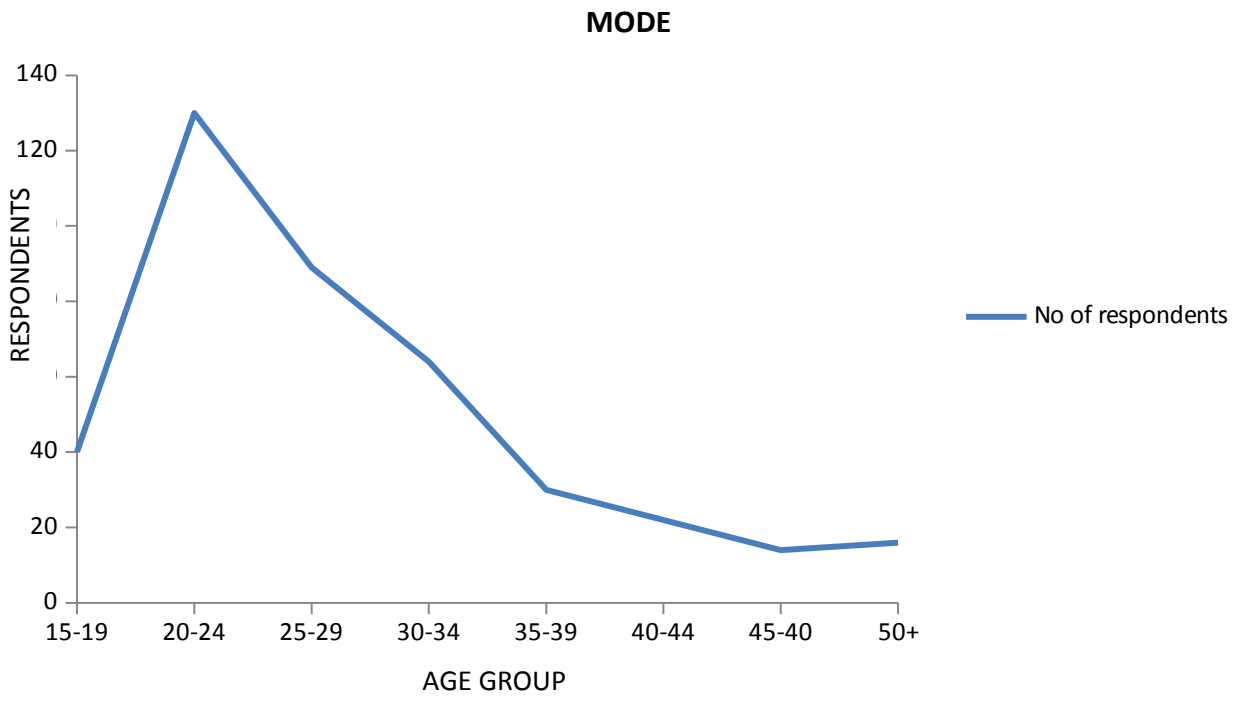
=23.43

Interpretation

Most of the respondents on the role of the community on HIV/AIDS from Namilama were aged 23 years .

Example 2

MODE DIAGRAMATICALLY



The Mode Value Is 23 As Per the Red Line.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

MONTH	2001	2002	MATCT
JANUARY	119	131	1429 119
FEBRUARY	126	136	1427 245
MARCH	125	132	1424 370
APRIL	111	135	1441 481
MAY	127	125	1415 608
JUNE	119	130	1428 727
JULY	125	130	1422 852
AUGUST	106	131	1442 958
SEPTEMBER	117	126	1426 1075
OCTOBER	119	132	1430 1194
NOVEMBER	98	135	1454 1292
DECEMBER	125	135	1427 1417
TOTAL	1417	1578	

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

MONTH	2002	2001	CT	MAT
JANUARY	533	317	317	4217
FEB	487	314	631	4165
MAR	399	283	914	
APRIL	586	219	1133	
MAY	594	111	1244	
JUNE	613	483	1727	
JULY	670	214	1941	
AUG	532	251	2192	
SEP	504	196	2388	
OCT	607	900	3288	
NOV	713	433	3721	
DEC	642	280	4001	
TOTAL	6871	4001		

GRAPH

QUARTILES

Are measures that divides a distribution into four equal parts

Types of data

1. Grouped data
2. Ungrouped data

Quarters

1st quarter 2nd quarter 3rd quarter 4th quarter

Abbreviation Q_1 Q_2 Q_3 Q_4

QUARTERS

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

1st quarter (Q_1) ,2nd quarter(Q_2), 3rd quarter(Q_3), 4th quarter (Q_4

Quartile in grouped data

Procedure

- Get the data
- Determine the quartile 1 position $\frac{N+1}{4}$
- $Q1 = \frac{N+1}{4}$ N—No. of items
- Arrange the items in order of size or magnitude e.g from smallest to highest.
- Identify the quartile position using the computed position values.
- Establish the values in the quartile position. Remember the odd and even distribution i.e odd distribution gives an actual value in the distribution while even distribution requires an approximation between the two values.
- The item(value) that is located in the quartile position in the quartile value.

Example

The following weights collected from patients attending ophthalmology clinic(92,71,60,70,75,50,71,74,60,74,69,63,62,50,71,80,80)kegs'.

1. $Q_1 = \frac{17+1}{4}$
2. Arrange items
   50,50,60,60,62,63,69,70,71,71,71,74,74,75,80,80,92
3. Determine the position 4.5
   $Q_1 = \frac{60+62}{2} = \frac{122}{2} = 61$

= 61KGS

$Q_3 = 3\frac{(17+1)}{4} = 45*3 = 13.5$

$= \frac{74+75}{2} = 74.5$KGS

<u>QUARTILES IN GROUPED DATA</u>

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

- Computation
- Diagrammatically

COMPUTATION

- Obtain the grouped data
- Determine the number of items referred to as sum of frequency
- Determine the quartile position by dividing the sum of the frequency by 4 and multiply by the desired quartile.

  $Q_1 = {}^{1}$_____Quartile position

  $Q_3 = 3$

- Locate the quartile position on the distribution table
- Substitute the quartile computation formula

  Quartile group = quartile position-CT class interval

  Quartile class frequency

  $Q_1 = L_1 +$

EXAMPLE

The following data shows respondents in a research process.

AGES	FREQUENCY	C.T
15-19	40	40
20-24	130	170
25-29	59	259
30-34	64	323

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

35-39	30	353
40-44	22	375
45-49	14	389
50+	16	405
TOTAL	405	

$Q_1 = L_1 +$

$Q_1 = 1 = 101.25$ position

$Q_1 = 20+5$

$=205$

$=22.4$ yrs

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

$Q_3 = {}^{3}$

$= 101.253$

$= 303.75$

$Q_3 = L_1 +$

$= 30 + {}^{5}$

$= 30 + {}^{5}$

$= 30 + {}^{5}$

$= 30 + 3.476$

$= 33.47$

$= 33.5$yrs

DECILES

When the distributions are divided into 10 equal parts are called deciles e.g.D_1, D2 ,D_3 ,D_4 , equal to median.

D_1 position = (ungrouped)

D_7 ^{7} (grouped)

PERCENTILES

When the distribution are divided into 100 equal parts are called percentiles

P_1 ,P_2 ,P_{25} ,P_{30} , P_{100}, etc

$P_{25} = Q_1$

P_{50}=median

P_1=(ungrouped)

P_{49} =49 (grouped)

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Example

The following are the ages of patients admitted at molo hospital on weekly basis

32,63,33,57,35,54,38,53,42,51,48,28,28,28,26,13,42,34,55,67,23,22,24,21,27,17,25,23,23,22,21,17,13,30,14,29,16,42,51,42,53,38,54,35,57,53,63,32,43,21,24,22,23,61,55,34,42,13,28,28,48,28,16,29,14,30,13,17,21,22,23,25,21,43,46,61,53,25,22,22,33,57,23,23,35,54,38,28,48,42,16,26,28,14,20.

a) find the decile;

i. D_3
ii. D_5
iii. D_7

b)percentile

I. P_{15}
II. P_{25}
III. P_{50}
IV. P_{75}

Example

The following data was collected on period of stay in kibera estate

Period in yrs

Period in years	residence
1	80
1 - 5	146
5 - 10	81
above 10	98

Determine deciles and percentiles

a) find the decile;

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

iv. D_3
v. D_5
vi. D_7

b)percentile

V. P_{15}
VI. P_{25}
VII. P_{50}
VIII. P_{75}

SIGNS AND SYMBOLS.

N / n -Number of items

∑ -Sum

f -Frequency

X -Individual item

- Arithmetic mean

i-Class interval

Z – Mode

Q – Quartiles (Q_1 , Q_3)

D -Deciles (D_1 ,D_7)

P – Percentiles (P_2 ,P_5)

L – Lower class limit (L_1)

CT –Cumulative frequency

d –deviation

d^1-deviation from assumed mean

MD median deviation

MEASURES OF DISPERSION / VARIABILITY

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

These are measures that help us to understand how individual items in any given distribution are distributed or positioned. This is done by comparing individual items, measures of central tendency and the total number of items in then distribution. All these are worked out leading to certain indicators that show variability / dispersion .Some of these measures include;

1. Range –is the difference between the highest and the lowest value in any given distribution.

These usually gives a value that can dictate (point to us) how close the items in a distribution were ie A range of higher values with fewer items indicate that the items are likely to be far from each other . A range of lower values with many items indicates that items are likely to be close to each other.

Example one

Mbagathi ANC weighed 40 clients below 50kgs and 90kgs while Maralal district hospital weighed 39 clients between 49kgs and 69kgs

Mbagathi -90 -50 =40

Maralal 69 – 49 =20

Interpretation

Its likely that the weights of clients inMaralal are closely similar as compared to Mbagathi

2. Mean deviation – this is a measure of variation which seeks to show a common value of the individual item s variations from a central value usually the arithmetic mean . Therefore it gains its name from the measure of the central tendency .
   It may also be referred to as average deviation.
   Therefore to calculate the mean deviation one would require to obtain all the deviations of the difference of the individual items in the distribution from the arithmetic mean
   Mean deviation is abbreviated as (MD) while deviation abbreviated as (d)

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Computation of mean deviation honors the category of data in the distribution; however, the principle is still the same.

a) Mean deviation of ungrouped data

Procedures

- **i.** Obtain the data
- **ii.** Calculate the arithmetic mean
- **iii.** Obtain the deviation by subtracting the arithmetic mean from the individual item
- **iv.** Obtain the sum of the deviation while ignoring the accompanying mathematical signs
- **v.** Divide the sum of the deviations by the total number of items abbreviated as

The resultant value after the division is the mean deviation for that particular distribution . This is formulated as follows

Mean deviation =

MD =

Example

The following weights were collected from patients attending ophthalmology clinic

92 ,71 ,60,70,75,50,71,74,60,74,69,63,62,50,71,80,80 kgs

Calculate the mean deviation

MD =

=

=8.1

b) Mean deviation in grouped data

Procedures

- **i.** Obtain the data
- **ii.** Compute the arithmetic mean
- **iii.** Workout the deviations (usually abbreviated as d) by subtracting arithmetic mean from the midpoint of each class

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

iv. Multiply the deviation by their respective frequency , this is abbreviated as fd

v. Sum up the product of the frequency and deviations disregarding the mathematical sign , this is abbreviated as

vi. Divide the sum of the frequency and deviations by the sum of the frequency in the distribution , this is abbreviated as

vii. The resultant answer upon division is the mean deviation for the distribution

Therefore , formula for mean deviation in grouped frequency is;

Mean deviation =

MD=

3. Quartile deviation – this is the measure of dispersion that indicate the relationship between quartile one and quartile three(Q_1& Q_3) in any given distribution

It is abbreviated as QD and it seeks to show the item value that gives the relationship between the lower quartile (Q_1) and the upper quartile (Q_3)

Its also referred to as interquartile range

To compute quartile deviation , the following steps are observed;

Step 1

Determine the lower quartile (Q_1) and the upper quartile (Q_3)

Step 2

Subtract the lower quartile value from the upper quartile value (Q3 - Q1)

Step 3

Divide the results answer by 2 =

The resultant answer is the QD for the distribution

This can be summed up as

Quartile deviation =

QD =

Example

Adultsresidence distribution among a research sample in Kenya

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Age 5- yrs group	Urban	Rural	Urban	Rural
15-19	90	174	90	174
20-24	123	133	213	307
25-29	96	107	309	414
30-34	69	78	378	492
35-39	67	95	445	517
40-44	22	75	467	662
45-49	27	70	494	732
Totals	494	732		

Compute;

Quartile 1 position

Quartile 3 for urban

Interpretation

Variation in age among the urban residence was less compared to rural residence .

Note :

Its worthy noting that the quartile deviation indicates uniformity in items hence when the quartile deviation is very low , It indicates that there is high uniformity among the values (items). Thus the items did not differ or vary very much and when the quartile deviations is very high it did not indicate that there is low uniformity among the items or values . Thus the items varied greatly or highly .

Assignment

STUDENT SCORE IN THREE SUBJECTS

COMPUTER SCINCE	HOME SCINCE	MATHS
62	53	85
57	76	90

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

75	75	99
72	51	50
56	57	85
77	62	65
67	58	69
59	51	53
72	56	69
60	75	50
56	51	62
63	37	98
53	50	99
67	62	91
78	71	52
74	50	51

Compute Q.D for student's score in

i. Computer science
ii. Home science
iii. Maths

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

The table below shows educational status among a selected research sample

Age -5 year group	No .education	primary	secondary
15-19	38	161	65
20-24	45	130	79
25-29	43	102	56
30-34	32	70	40
35-39	75	57	27
40-44	54	30	13
45-49	60	32	4
Total	347	582	284

Compute the

a) mean deviation
b) quartile deviation
c) interpret your answers

VARIANCE

Variance is the name given to the square of the standard deviation . ie variance = S^2 . It is important because variance can be added . For instance , if two independent distributions had variances of S_1^2 and S_2^2 respectively the variance of the two distributions combined would be $S_1^2 + S_2^2$.

Is a kind of dispersion which seeks to overcome the ignored mathematical principles

Ungrouped –V $=^2$

Grouped - ^{2}

SD =

Standard deviation symbolize

Is found by adding the squares of the deviations of the individual value from the mean of the distribution , dividing this sum by the number of items in the

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

distribution , and then finding the square root of the quotient . Algebraically, this can be shown as

SD = SD $=^2$ -ungrouped

The more the values of individual items differ from the mean , the greater will be the square of these differences and therefore the greater sum of the squares . And the greater this sum , of course the larger will S be , Hence , the greater the dispersion , the larger the standard deviation will be .

NOTE: if there is no dispersion at all , ie all values are the same , the S will work at zero

Interpretation

The individual performance in mathematics varied greatly compared to home science and computer science

The individual performance in computer science was better compared to home science and maths

STUDENT SCORE IN THREE SUBJECTS

COMPUTER SCINCE	HOME SCINCE	MATHS
62	53	85
57	76	90
75	75	99
72	51	50
56	57	85
77	62	65
67	58	69
59	51	53
72	56	69
60	75	50
56	51	62
63	37	98
53	50	99
67	62	91
78	71	52

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

74	50	51

Compute the

a. arithmetic mean
b. variance
c. standard deviation for the three subjects ie home science , computer science and maths

VARIANCE –is a measure of variation which seeks to overcome the act of ignoring the mathematical sign undertaken in the mean deviation computation . This makes it better measure of variation hence its name variance

Procedure

1. obtain the data in a given distribution
2. understand the data , the category in which it exists and determine the method to be followed in computation
3. work out the arithmetic mean for the distribution)
4. obtain the deviations by subtracting the mean from every item (ungrouped) or midpoint in grouped abbreviated as (d)
5. square each deviation as obtained abbreviated as (d^2)
6. multiply the squared deviations by the number of items or frequency in each class abbreviated as (fd^2)
7. obtain the sum of the number multiplied by the squared deviations abbreviated as $\sum d^2$ or $\sum fd^2$
8. divide the resultant sum by the total number of items or sum of frequency abbreviated as ^{2} OR ^{2}
   The resultant answer is the variance of the distribution abbreviated as S^2
   This is formulated as follows =–ungrouped variance

$$S^2 =^2$$

grouped variance

$$S^2 =^2$$

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

STARNDARD DEVIATIONS

This is a measure of variation or dispersion that is said to be the best in determining the spread of items in any given distribution

This is because it involves each item in the distribution individuals and its deviation

It is also referred to as root squared mean deviation . This is because its obtained as the square root of the mean of the square deviation of the arithmetic mean

Usually , the squared deviations are sum up before the standard deviation is obtained

The standard deviation is abbreviated as SD or symbolized as

Its also referred to as the square root of the variance hence abbreviated as ^{2}

PROCEDURE

1 Observe all the steps in obtaining variance

2 work out the square root of the variance abbreviated as ^{2}

3 The resultant answer is the standard deviation for the distribution .This can be formulated as follows

^{2} or $= {}^{2}$ or SD^{2}

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

STANDARD DEVIATION INTERPRETATION

The higher the value of standard deviation the greater the level of variation or dispersion in the distribution . This is also referred to as variability level of that particular distribution. Thus it would mean that the magnitude of deviation of values in the distribution from the arithmetic mean is greater . A smaller standard deviation value means that there was low variability level of values (that is the distribution) from the arithmetic mean

Thus a higher degree of uniformity of value or items in that particular distribution . Therefore the standard deviation can be used in deciding or making a choice between two or more distributions especially when they may have the same values or their measure of central tendency (arithmetic mean)

In certain cases distributions with smaller standard deviation would mean that particular distribution is the one with the most dependable or representative arithmetic mean .

This makes standard deviation measurement a very useful indicator in checking the representativeness of arithmetic mean in a given distribution .

COEFFICIENCY OF VARIATION

This is a measure of variation that seeks to bring a clear difference (level) of variation in two or more distributions

It involves the arithmetic mean and standard deviation and is considered as the best relative measure of variation

It's also referred as the coefficient of standard deviation (SD).It is also abbreviated as(cv).

PROCEDURE TO OBTAIN

1) **O**btain the data
2) Calculate the arithmetic mean and standard deviation for the distr4ibution abbreviated as and SD
3) Divide the standard value by the in the distribution
4) Multiply the resultant answer by 100 to give a percentage expression .

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

All this are expressed as follows.

Coefficient variation =

CV=

The resultant is a percentage expression

Example 1

Population distribution in coast for 1989 and 1999.

		SD	CV
1989	21.3	17	99.8
1999	21.7	11.7	76.95

1989=

=99.8%

1999= =76.95%

EXAMPLE 2

The table below shows educational status among a selected research sample

Age -5 year group	No .education	primary	secondary
15-19	38	161	65
20-24	45	130	79
25-29	43	102	56
30-34	32	70	40
35-39	75	57	27
40-44	54	30	13
45-49	60	32	4
Total	347	582	284

Calculate;

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

I. Arithmetic mean
II. Variance
III. Standard deviation

VI. Work out co-efficiency of variation for education status of primary and secondary.

The following data refers to geriatrics clinic attendance and weights of patients in these clinics(weight in kg)

Marsabit	Samburu
65	57
70	49
66	46
59	79
57	72
62	54
66	66
71	65
58	63
67	76
61	45
68	95
63	62
65	68
71	50
69	53
67	58
74	65
72	69
60	72

Calculate the following for samburu and marsabit

I. Arithmetic mean
II. Quartile deviations
III. Standard deviation
IV. Co-efficiency of variation

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

SKEWNESS

This is a comparable characteristic which helps us to understand distributions further or more this is especially where one has similar mean and standard deviation in distribution but the end diagrammatic view appearance is different among the distribution.

Therefore a measure of skewness indicates the difference between the manner in which items are distributed or spread in particular groups of items in a distribution in relation to the normal distribution.

The presentation showing the difference in appearance are referred as asymmetrical and the one which is normal (no skewedness) is referred to as symmetrical.

The difference in spread from the normal may be in the lower values or in the higher values. Where the spread extents into the higher values, the distribution curve on presentation tends to be higher on the right hand side of the x –axis. Such a distribution is referred to as positively skewed .where the spread extents more in the lower values of the distribution, the distribution curve when presentment on the x-axis tents to be higher than left part of the x-axis. Such distribution is referred to us negatively skewed

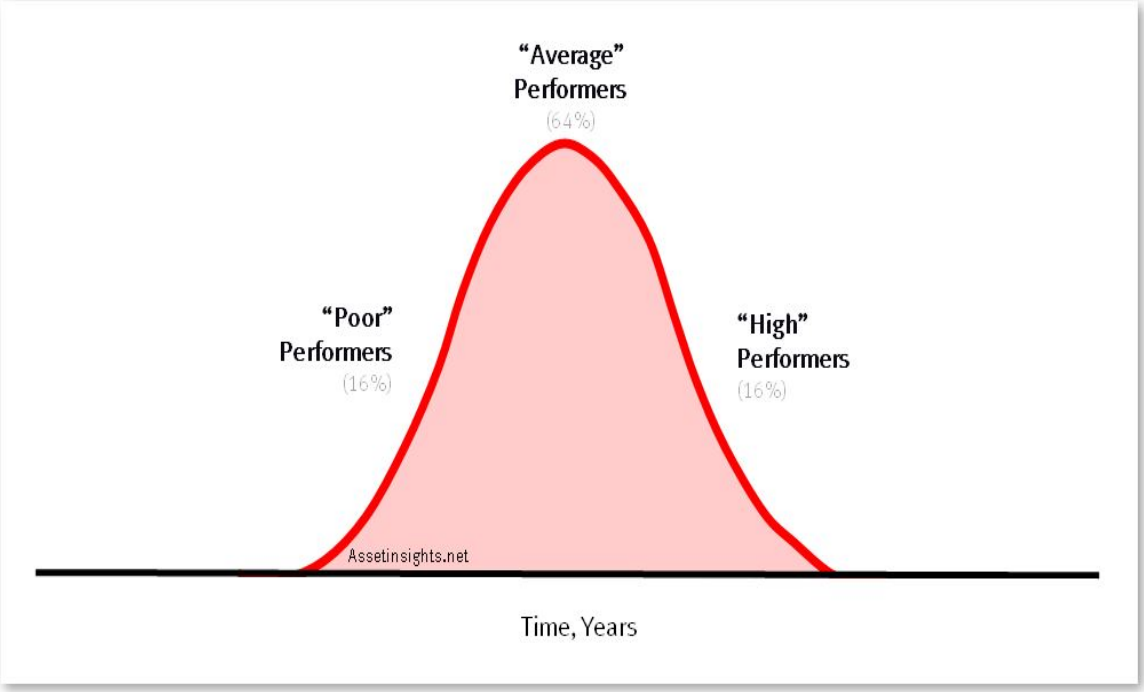
The three common measures of central tendency arithmetic mean, median and mode also help us to describe skewness easily and to distinguish their differences in appearance

These are simplified in appearance as follows;

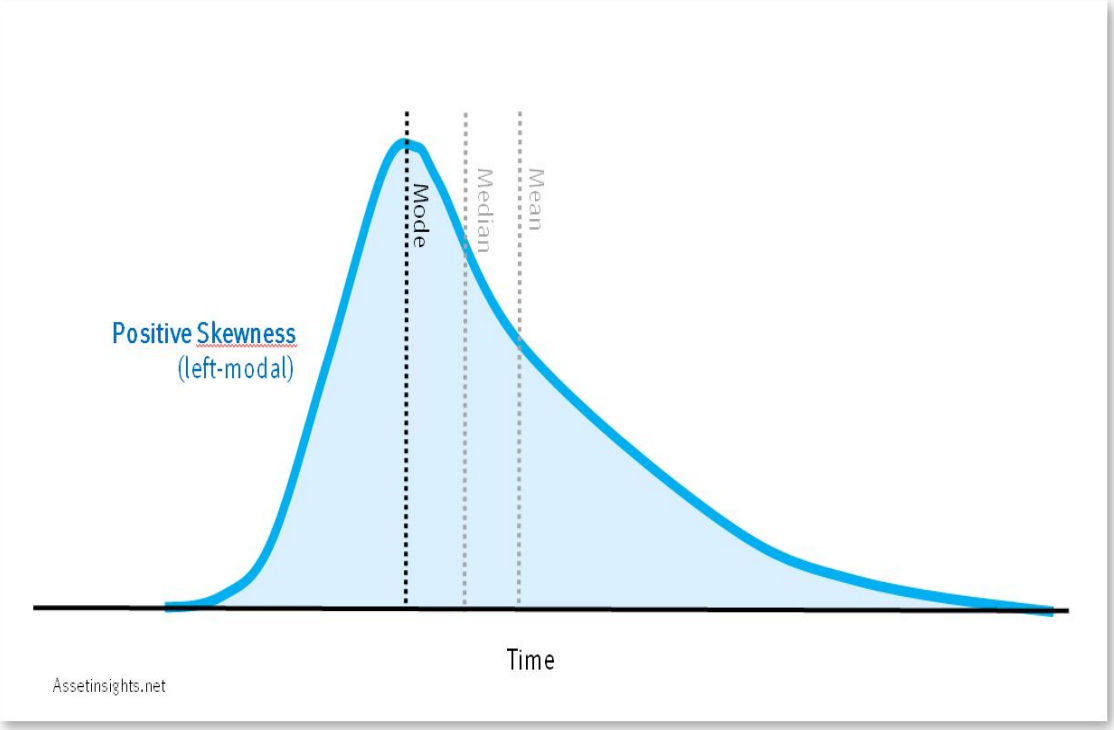
.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

1. SYMMETRICAL / NORMAL DISTRIBUTION



2. POSITIVELY SKEWED DISTRIBUTION

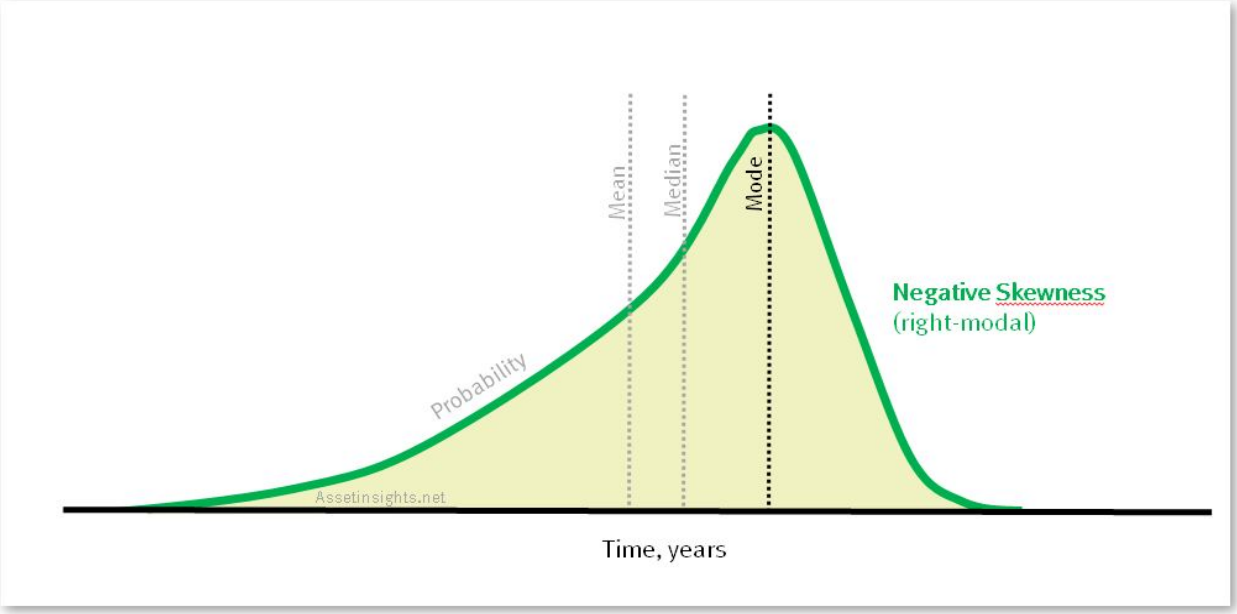


.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

3. NEGATIVELY SKEWED DISTRIBUTION



METHODS OF CALCULATING SKEWNESS

As mentioned above there are three common measures of central tendency also helps us to describe skewness and determination of its direction or category of skewness

There are many methods and procedures of determining skewness in any given distribution named or based on the names of people who first suggested and proposed them and they are in determining skewness. Some of the commonly used methods include ;

1. PEARSONIAN MEASURE OF SKEWNESS

As the name suggests this was proposed by professor KaarlPearsons . He who a great british by biometrician and statistician from 1857-1936.

Its based on the fact that mode and mean have a difference in any skewed distribution .This difference is then divided by the standard deviation of the same distribution .

Inn determining skewness , this method is formulated as follows ;

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Skewness =

Sk=

Most values of skewness obtained are between minus one and positive one. Although in certain circumstances:

This value may be up to minus or positive three but rarely higher or beyond this values e.g

Example 1

Patient's weight in a special clinic were established as follows ;

Kgs	No. of pts(f)	m.p(x)	fx	ct	d	D^2	Fd^2
54	40	52	2080	40	-13.8	189.1	756.4
55-59	70	57	3990	110	-8.8	76.6	536.2
60-64	130	62	8060	240	-3.8	14.1	183.3
65-69	80	67	5360	320	1.3	1.6	128
70-74	80	72	5760	400	6.3	39.1	3128
75-79	50	77	3850	450	11.3	126.6	6330
80+	30	82	2460	480	16.3	264.1	7923
Totals	480		31560		61.6	711.2	32268

Calculate the following

a) arithmetic mean
b) mode
c) state the modal class
d) standard deviation
e) variance
f) skewness

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

example 2

the following data was collected on adults referred to the laboratory for malaria parasites test on monthly basis in the kabsebeth hospital

Age	F	Ct	m.p (x)	Fx	d	D^2	Fd2
15-19	24	24	17	408	-14	196	4704
20-24	57	81	22	1254	-9	81	4617
25-29	75	156	27	2025	-4	16	1200
30-34	74	230	32	2368	1	1	74
35-39	67	297	37	2479	6	36	2412
40-44	32	329	42	1344	11	121	3872
45-49	32	351	47	1034	16	256	5632
Total	351			10912			22511

Compute the following using the table above

(a) arithmetic mean
(b) mode
(c) standard deviation
(d) skewness

its worthy nothing that in normal distribution (symmetric distribution) mean , mode , and median are equal hence upon application of the pearsonian formula the resultant answer will be zero which would mean that the distribution has no skewness .And upon computation where the resultant answer has appositive sign it denotes that the distribution is positively skewed . Which will mean that the spread is more in the higher values of that particular distribution , while where the resultant answer has a negative sign it denotes that the distribution is negatively skewed . Thus the spread extends more or exists more among the lower values or items .

the resultant numerical values also give the extend or level of the skewness ie where two distributions exists and one has a skewness values of + 0.65 and the one of + 0.85 the later (+0.85) is more skewed than the earlier distribution .

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Consequently , this formula gives both the direction and extends of skewness in any given distribution

2. MODIFIED PEARSONIAN MEASURE OF SKEWNESS

In certain distributions mode may not be exact especially in grouped frequency distribution . This formula is used on the averages of moderately skewed distribution whose relationship indicates that mode = 3 median -2 mean when this relationship is applied on the first formula of skewness measurement described and the formula is modified as follows ;

Skewness (sk) =3

Most values again are ± 1 although they vary between ± 3 and the resultant answer sign gives the direction and level of skewness as explained in the first formula . in most computation it is recommended that the first formula to be used in computing skewness for ungrouped distribution while the second for grouped frequency distribution

Example 1

Use the data below

The following data was collected as patients bp in the hypertension clinic

Bp	Pts (f)	Ct	m.p (x)	Fx	d	D^2	Fd^2
71-80	12	12	75.5	906	-37	1369	16428
81-90	18	30	85.5	1539	-26.5	702.25	12640.5
91-100	35	65	95.5	3342.5	-16.5	272.25	9528.75
101-110	42	107	105.5	4431	-6.5	42.25	1774.5
111-120	50	157	115.5	5775	3.5	12.25	612.5
121-130	45	202	125.5	5647.5	13.5	182.25	8201.25
131-140	30	232	135.5	4065	23.5	552.25	16567.7
141-150	8	240	145.5	1164	33.5	1122.25	8978
Total	240			26870			74731

Calculate the following

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

a) Mean
b) Mode
c) Median
d) Skewness

Example 2

Below are weights of 2002 diploma class

Class	F	Ct	Mp (x)	Fx	d	D^2	Fd^2
42-51	5	5	46.5	2325	-19.6	384.16	1920.8
52-61	12	17	56.5.	678	-9.6	92.16	1105.92
62-71	17	34	66.5	1130.5	0.4	0.16	2.27
72-81	12	46	76.5	918	10.4	108.16	1297.92
82-91	4	50	86.5	346	20.4	416.16	1664.64
Total	50			3305			5992

Compute

i.) Mean
ii.) Median
iii.) Skewness

3. BOWLEY'S MEASURE OF SKEWNESS

Formula

Sk =

This method was first proposed by professor Bowley's based on quartile with remembrance that symmetrical distribution have their first and third quartile situated equal distance from the median value which is also the second quartile value. This gives and impression that can be expressed as follows;

Q3-median=median =Q1

Or

Q3 +Q1-2 median =0

His principles therefore are that in asymmetrical distribution there will be a difference in these equations if the quartile is worked out. The different results

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

after computing the quartiles in the impression would indicate that either the 25% of items at the top or below the median are further from it. This impression can be expressed into a formula that can be used to measure skewness as follows;

SK =

The interpretation of the skewness resultant value will be as in the other measurements previously explained. The positive or negative skewness of up to ±3 signify variation in the first 25% item (negative) or the top 25% item (positive). It is also interpreted that if it is positive skewed the top 25% of the items in the distribution tended to be further from the median than the first 25% of the items. Thus the Q3 value will be further from median than the Q1 value. This would be vice versa for the negatively skewed distribution.

It is advisable that the skewness value obtained out of the two formulas should not be directly compared with each other since the numerical values have no direct relationship. However a direct common feature would be attained in a given distribution on direction and existence of skewness. It is rare to obtain different direction for the two formulas from the same distribution.

NOTE

It is worthy to remember that skewness gives us the direction in a distribution which has more values from a central measure and with more varied spread among the items. Thus the extend of spread among the items is more therefore effective interpretation should consider the measure of central tendency and the method of computation used.

Example

This sample data was collected from Namilama division during a research process on the role of the community on HIV/AIDS awareness . compute the skewness for the total and explain the outcome answer-meaning of the skewness

Age	male	female	Total	Ct
15-19	17	23	40	40
20-24	55	75	130	170
25-29	46	43	89	259
30-34	31	33	64	323

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

35-39	18	12	30	353
40-44	11	11	22	373
45-49	9	5	14	389
50+	15	1	16	405
Total			405	

NORMAL DISTRIBUTIONS

This is the most important of all statistical distribution analysis.This was just discovered by a French mathematician –Abraham Demoivre in 1733 and was rediscovered and applied to normal and natural social sciences by the French mathematician Pierre simondelaplace and the Germany mathematician and astronomer Karl Friedrick Gauss in the early C19th hence its also referred to as Gaussian distribution

Many phenomena closely follow or approximately follow normal distribution , this include ;

- ✓ Weight of persons
- ✓ Serum
- ✓ Cholesterol
- ✓ Life span of light bulbs
- ✓ Body temperature of health persons
- ✓ Size of oranges ,bananas etc

We have got also other phenomena that do not follow the normal distribution such as individual annual earnings , clinical laboratory readings of various specimen ie urea , magnesium etc

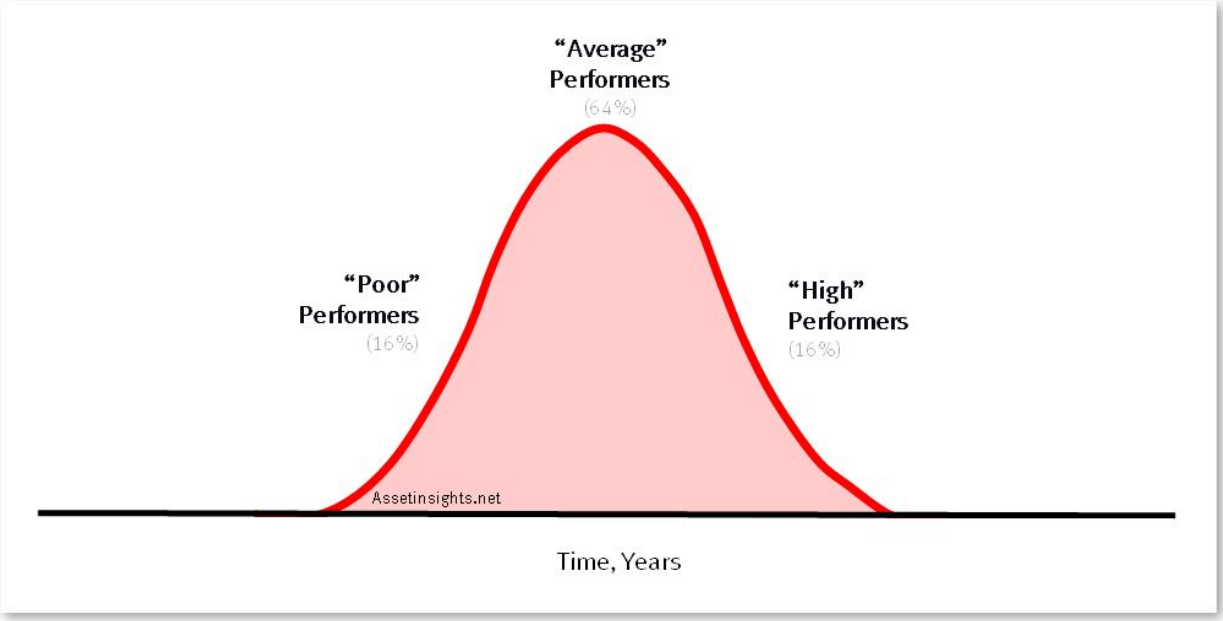
Normal distribution are also popular because they posess certain mathematical properties that make them attractive and easy to manipulate .

Its also based on the theory of methodology developed on assumption that certain data are distributed approximately on normal basis

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

NORMAL DISTRIBUTION CURVE



PROPERTIES OF NORMAL DISTRIBUTION

1. Bell –shaped curve upon diagrammatic presentation of the distribution on a curve
2. Have particular internal distribution for the area under the curve which is referred to as confidence limits . The confidence limit property relates the arithmetic mean and the standard deviation of distribution in such a way that;
   68.26% of items are included in arithmetic mean plus or minus one standard deviation value (68.26%= ±SD)

   Also 95.45% of the items are included in arithmetic mean plus or minus two standard deviation (95.45% = ± 2SD)
   In the same line 99.74% = ± 3SD)

3. The amount of area under the normal curve is directly proportional to the x-axis score

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

CORRELATION AND REGRESSION

These are measures which help us determine the existence of a relationship between two or more variables in a distribution

Also in prediction of possible value one of the related values are given

CORRELATION

Is a statistical term which indicates two quantities / distributions varying in such a way that movement or change in one is followed / accompanied by movement or change on the other .

Such quantities are said to be correlated

Example

- Fundal heights and gestation age
- Age of husband and age of wife
- Distance covered and speed used

CORRELATION ANALYSIS

Correlation analysis determines whether there is a statistically probable relationship between two variables .It does not presume as it does causal analysis ;that one variable actually is the agent that causes change to occur in another variable .It merely assesses the strength and direction n of an association between variables . Many empirical attempts to understand political phenomena

Causation is very difficult to assess using statistics of correlation .Usually, we try to define one variable as the independent or causal factor in a relationship and the other variable . For example :smoking tobacco(independent variable)is correlated and usually precedes getting cancer (dependent variable) or another example is us politics ;women are more likely than men to vote for the democratic party , gender would be independent variable and their vote for president (democrat or republican) would be the dependent variable .

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

The actual correlation factor in the 1996 elections was 24.This means that if we know the gender of a person ,we have a 24% vote better chance of predicting their vote than if we don't know their gender causes people to vote in different ways , it simply indicates that there is a relationship that may be causal in nature

Causation about correlational analysis correlation is not the same as cause . The fact that people who find jobs with health coverage are more likely to work steadily does not mean that providing coverage to low wage workers will necessarily improve retention

CORRELATION ANALYSIS

This is the attempt / consideration to measure / determine the degree of relationship between two or more variables that are correlated

A statistical device that helps in analyzing relationships between distribution / series

USES OF CORRELATION

- Determination if a relationship existence between two distribution / variable
- Measure relationship between two variables
- Test if the relationship is relevant or significant
- Establishment of causes and effects if there is any between the variables with relationship
- Estimation of values when one is known

TYPES OF CORRELATION

May be classified in different ways that include

1. Positive or negative
2. Simple ,partial and multiple
3. Linear and non linear

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

POSITIVE CORRELATION

Depends on direction of change on the variables . This may occur in two ways

-where one variable increase the other also increase

-where one decrease the other one also decrease

It is also referred to as direct correlation

Example

-variable x 10,12,15,18,or,80,70,60,40,40

-variable y 15,20,22,23,37,or,50,44,30,20,10

NEGATIVE CORRELATION

Variables vary in the positive direction such that where one variable increase the other one decreases and vice versa

Its also referred to as inverse

Example

-variable x 20,30,40,60,80,or 100,90,60,40,30

-variable y 40,30,22,15,10,or 10,20,30,40,30

SIMPLE , PARTIAL AND MULTIPLE CORRELATION

Simple –only two variables are studies as above

Partial and multiple – three or more variables are studied

Partial- more than two variables are reorganized but only two are considered to influence each other .The possible influence from the others are kept constant

Multiple – three or more variables are studies simultaneously ie a drug test on the station , samples , amount used and effectiveness

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

LINEAR AND NON LINEAR

These are correlation on the relation of change

The distinction between the two is based on the consistence of ratio change in the variables

Linear –is where change in one variable bear constant change in the other variable

Example

-variable x 10,20,30,40,50

-variable y 70,140,210,280,350

If the ratio is presented on the graph it gives a straight line

NON LINEAR

Non linear – is where amount of change does not bear constant ratio of change to the other variable

It is also referred to as curvilinear

Double ratio of change in one does not necessarily mean double ratio in the other variable

METHODS OF STUDYING CORRELATION

Diagrammatical

-scatter diagrams

-graphs

Mathematical computation

-karlpearson's coefficient of correlation

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

-Concurrent deviation

Interpretation same

- Perfect, high degree , low degree positive relation
- Perfect, high degree, low degree negative relation

REGRESSION

The act of returning back,first used by sir Francis Calton in 1977 while studying the relationships between heights of fathers and sons published in the paper regression towards mediocrity in hereditary stature .It showed that tall fathers tend to have short sons while short fathers have short sons .

Measure of average relation between two variables in terms of the original unit of data or variables

REGRESSION ANALYSIS

Once a relationship has been established there would be a possibility of estimation or prediction of values of one set where the other set is known and this is referred to as regression analysis

It also refer to the methodology in which estimates are made of values of variables from knowledge of one or two of the other variables

Also measurements of errors are involved in this estimation process .

REGRESSION STUDY

This is done by drawing lines which are referred to as regression lines

USES

i. Estimation of independent and dependent variables
ii. Obtaining measure of error involved in using lines of regression as estimation basis
iii. Can be used to calculate correlation coefficient

STATISTICAL ANALYSIS

.

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)

Is concerned with collection, organization and interpretation of data according to well defined procedure.

It is the heart of every scientific discipline

Is a methodology of systematically analyzing and drawing conclusion on things studied

STUDY AREAS

- Behavioral and biological characteristics of human being
- Aspects of plants and animal life
- Dynamics of physical environment
- Interactions of any of these dynamics

GOALS

In broad sense it is the process of drawing conclusions and understanding more about the sources of our data.

This manipulation and organization has basic objectives as follows ;

-reduction of data

-procedures of starting degree of confidence in the accuracy of the measures

-identification of associations

.

This document is available on studocu

Downloaded by Patrick Ngobiro (pngobiro@gmail.com)